Pan-Cancer Hierarchical Spike-and-Slab — Replication & Validation

Pan-Cancer Hierarchical Spike-and-Slab — Replication & Structural-Prior Validation

A pre-registered held-out validation in which structural priors derived from external biological hierarchies replace the per-cancer fitted PIPs for four covariates of the Lock-lab pan-cancer survival spike-and-slab model. The pre-registered decision rule fires **OUTPERFORMS** for the primary specification.

TL;DR

model	held-out total LPPD	Δ vs baseline	bootstrap 95% CI	verdict
baseline (Samorodnitsky/Hoadley/Lock 2022, fit on 80% training)	−1038.69	—	—	—
primary structural-prior	**−1032.43**	**+6.26 nats**	**[+0.34, +12.29]**	**OUTPERFORMS**
secondary structural-prior	−1034.36	+4.33 nats	[−1.86, +10.97]	inconclusive

Held-out test set: 1,362 patients across 29 cancer types, stratified random 20% per cohort, seed 42. Four chains × 100,000 Gibbs iters per model, 50k burn, 500 thinned posterior samples for LPPD evaluation, B=1000 patient-level bootstrap.

Decision rule (locked before any validation Gibbs run, see `PRE_REGISTRATION.md`): a model "outperforms" the baseline if held-out LPPD exceeds the baseline by > +2 nats *and* the bootstrap 95% CI on the difference excludes zero. Primary specification meets both criteria.

The structural priors replace 116 free per-cancer beta parameters with 31 group-shared betas across the four target covariates (a 73% reduction). Held-out predictive performance improves nonetheless — consistent with productive regularization, not flexibility loss.

Provenance

The replicated method:

> *Samorodnitsky, S., Hoadley, K. A., & Lock, E. F. (2022). A Hierarchical Spike-and-Slab Model for Pan-Cancer Survival Using Pan-Omic Data. BMC Bioinformatics 23(1):235. Source: https://github.com/sarahsamorodnitsky/HierarchicalSS_PanCanPanOmics*

The BIDIFAC+ inputs:

> *Park, J., Lock, E. F., & Hoadley, K. A. (2022). Bidimensional Linked Matrix Factorization for Pan-Omics Pan-Cancer Analysis. Annals of Applied Statistics 16(1):484-501.*

The 29 TCGA cancer types and pre-computed `XYC_V2_WithAge_StandardizedPredictors.rda` inputs come directly from the upstream repository. No re-fit of BIDIFAC+ was performed.

What's in this repo

```
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├── README.md                                ← this file
├── README.pdf                               ← printable version
├── LICENSE                                  ← MIT (this work) + upstream attribution
├── PRE_REGISTRATION.md                      ← Paper 1 — locked decision rules
├── PRE_REGISTRATION_PAPER2.md               ← Paper 2 — operator-composition extension (pre-r
├── EXTENSION_VALIDATION_RESULTS.md          ← Paper 2 — 18-spec verdict + per-spec table
├── REPLICATION_RESULTS.md                   ← baseline replication numbers
├── SCREENING_SUMMARY.md                     ← screening pass output
├── FALSIFICATION_REPORT.md                  ← 6-check adversarial audit of screening
├── VALIDATION_RESULTS.md                    ← pre-registered held-out validation (Paper 1)
├── WRONGLY_GROUPED_DIAGNOSTIC.md            ← within-group sign-agreement diagnostic (Paper 2
├── COV_10_1_INVESTIGATION.md                ← 3-cov sensitivity test on Paper 1 primary (Read
├── FINER_RESOLUTION_RESULTS.md              ← cov 8.2 tissue-within-group ANOVA (discovery, A
├── reference/
│   └── cancer_type_hierarchies_2026-05-09.md ← 29 cancers × 5 hierarchies × granularities
├── scripts/
│   ├── replication/                         ← R Gibbs runner + diagnostics
│   ├── screening/                           ← Python ANOVA + falsification cascade
│   └── validation/                          ← R modified-Gibbs + held-out evaluation
├── results/
│   ├── replication/                         ← per-cancer × per-cov posterior table + diagnost
│   ├── screening/                           ← screening + 6 falsification CSVs
│   └── validation/                          ← LPPD, bootstrap, per-cancer, convergence
└── data/
    └── README.md                            ← pointers to XYC, TCGA, Zenodo (chain RDAs)
```

The chain RDA files (12 chains × ~580 MB = ~7 GB total) are **not in this repository**. They will be deposited at Zenodo with a citable DOI; the deposit URL goes here:

> ***Zenodo DOI: TBD*** *(will be filled in after the public release of this repo)*

Without the chain RDAs, downstream analyses (convergence, held-out LPPD, bootstrap) cannot be recomputed from saved posteriors directly. The full pipeline can still be re-run from scratch using the scripts here in ~2 hours wall on a modern workstation. See `data/README.md`.

What was done, briefly

1. Replication baseline

Four-chain replication of `HierarchicalLogNormalSpikeSlab` (the upstream sampler, unmodified) on the full pre-computed XYC inputs. R-hat max 1.0152 across 137 monitored params; per-chain σ^2 posteriors agree to 3 decimals. Cancer-specific intercepts span THCA 10.28 (≈80 yr at standardized predictors=0) to MESO 6.29 (≈1.5 yr) — clinically expected order. Pan-cancer age effect $\tilde{\beta}_{0.5} = -0.316$ (PIP 0.91). See `REPLICATION_RESULTS.md`.

2. Structural-hierarchy screening

ANOVA of per-cancer PIPs against five candidate hierarchies (tissue origin, epithelial classification, embryonic germ layer, sex composition, Hoadley 2018 supercluster/iCluster) at multiple granularities each. 636 tests total across 68 BIDIFAC+ covariates. 16 nominal hits at $\eta^2 > 0.30$ with $p < 0.05$. See `SCREENING_SUMMARY.md`.

3. Falsification cascade (six adversarial checks)

1. Permutation null on η^2 (B=1000): observed at chance rate
2. High-resolution perm on survivors (B=10000): 0/16 pass Bonferroni
3. BIDIFAC+ block-presence confound: Hoadley signal ρ=+0.60 with block presence (circular)
4. Granularity-nesting null on Spearman ρ: tissue origin marginal (p=0.042), all others fail
5. Cohort-size partialing: most survivors lose < 0.06 η^2, one loses 0.15
6. Independence: 16 survivors collapse to 5 effective dimensions; 32/120 pairs Jaccard > 0.5

No single (cov, hierarchy, granularity) cell is publishable as standalone structural evidence at conventional rigor. See `FALSIFICATION_REPORT.md`.

4. Pre-registered validation

The screening result motivated a different question: do the four largest-effect-size candidates *jointly* improve held-out predictive performance when used as structural priors, even though no single candidate is individually Bonferroni-significant? Pre-registered in `PRE_REGISTRATION.md`, ran the modified Gibbs sampler at `scripts/validation/sampler_structural_prior.R`, evaluated held-out LPPD on the 20% test set with B=1000 bootstrap CI. Result: primary specification outperforms baseline (Δ=+6.26 nats, CI excludes zero). See `VALIDATION_RESULTS.md`.

5. Follow-up investigations (post-shipping)

Three diagnostic / sensitivity analyses, all data-only reports without disposition recommendations:

- **`WRONGLY_GROUPED_DIAGNOSTIC.md`** — within-group sign-agreement scan of per-cancer baseline betas across all candidate (cov × hierarchy) cells used in Paper 1 and Paper 2. Two cells flag (both cov 10.1 at epi/2a and germ/3b); cov 8.2 is clean across all four candidate hierarchies. Prerequisite (b) of the Paper 2 pre-registration.
- **`COV_10_1_INVESTIGATION.md`** — refit Paper 1 primary on three covariates (drop cov 10.1) to test whether cov 10.1's contribution matters. 3-cov vs 4-cov $\Delta = -1.42$ nats, bootstrap CI includes zero (Reading A holds). The 3-cov spec alone has a CI that includes zero against baseline, so cov 10.1 is the marginal covariate carrying the 4-cov result across the CI-excludes-zero threshold.
- **`FINER_RESOLUTION_RESULTS.md`** — ANOVA + permutation null test of whether tissue-within-group or hybrid tissue+subtype labels organize cov 8.2 residuals beyond the coarse epi/2a 3-group prior used in Paper 1. Epithelial group n=21, η^2=0.608 vs perm null median 0.381, p_emp=0.117. Directional consistency but not certified at this n.

How to reproduce

Two paths:

Fast path — verify reported numbers from CSV outputs

The small CSV outputs in `results/` are sufficient to regenerate every table in the four reports without rerunning any Gibbs sampler. Inspect `results/validation/loglik_summary.csv`, `bootstrap_ci_summary.csv`, `per_cancer_loglik_wide.csv`, etc. directly.

Full path — re-fit everything from raw data

~2 hours wall on a modern workstation, four parallel chains per Gibbs run, three Gibbs runs sequentially. Step-by-step instructions in `data/README.md`. Briefly:

```
# 1. Pre-computed XYC inputs (clone upstream repo)
git clone https://github.com/sarahsamorodnitsky/HierarchicalSS_PanCanPanOmics.git

# 2. Replication baseline (4 chains × 100k iters; ~36 min wall)
for c in 1 2 3 4; do
  Rscript scripts/replication/run_gibbs_chain.R $c &
done
wait
Rscript scripts/replication/diagnostics.R
Rscript scripts/replication/per_cancer_summary.R

# 3. Screening + 6 falsification checks (~30 min)
python scripts/screening/hierarchies_long.py
python scripts/screening/run_screening.py
python scripts/screening/falsify_0{1,2,3,4,5,6}_*.py
python scripts/screening/synthesize_summary.py

# 4. Pre-registered validation (3 Gibbs runs sequentially × 4 parallel chains; ~2 h)
Rscript scripts/validation/01_build_split_and_targets.R
for spec in baseline_train primary secondary; do
  for c in 1 2 3 4; do
    Rscript scripts/validation/run_chain.R $spec $c &
  done
  wait
done
Rscript scripts/validation/05_convergence.R       # halt if R-hat > 1.05
Rscript scripts/validation/02_compute_held_out_loglik.R
Rscript scripts/validation/04_per_cancer_breakdown.R
Rscript scripts/validation/03_bootstrap_ci.R
```

Software: R 4.6.0 with `posterior` 1.7.0 + the upstream BSFP dependencies; Python 3.12 with pandas 2.3.3, scipy 1.17.1, numpy 2.4.4. The `infinitefactor` package was archived from CRAN on 2025-12-25 and the upstream code uses it indirectly through BSFP; a pure-R port of the relevant functions is documented in the validation pipeline if needed.

Random seeds are fixed throughout: chains 1–4 use seeds 20260509–20260512; the held-out split uses seed 42; bootstrap uses 20260510; permutation nulls use 20260509 and 20260511.

Paper 2 — operator-composition extension

A separate pre-registration covers a methodology-extension question: do operator compositions (combinations of structural priors across hierarchies) recover patterns single-operator priors miss? The specification list (3 covariates × 3 composition rules × 2 hierarchy pairs = 18 Gibbs specs), decision rules, and Bonferroni-corrected validation criteria are locked at `PRE_REGISTRATION_PAPER2.md`.

Compute has run. All 18 specs fit (4 chains × 100k iters each, 18/18 converged at R-hat ≤ 1.05). Per-spec verdicts under the pre-registered decision rule (≥ +2 nats Δ vs Paper 1 anchor, CI excludes 0, Bonferroni $p < 0.00278$) are reported in `EXTENSION_VALIDATION_RESULTS.md`. The secondary diagnostic anchor (vs baseline-on-train, no structural priors) is included in the same table.

Caveats

- **Validation tests four pre-specified covariates only**, not all 68. The result does not generalize to the full covariate set.
- **Single 80/20 split**, not k-fold cross-validation. Tighter CIs and per-cancer subgroup tests would require k-fold.
- **Tissue origin** is not used in any validation specification. It escapes the BIDIFAC+ block-presence confound but only marginally beats the granularity-nesting null. A tissue-origin spec would need a separate pre-registration and held-out test.
- **KIRP and STAD** lose under both validation specifications. The gain is heterogeneous; understanding why these cancers worsen is a candidate follow-up.
- **n=29** is small. The screening was demonstrably underpowered for 319 simultaneous tests at Bonferroni rigor. The validation succeeded by testing joint predictive contribution rather than per-cell significance.

Citing

This repository (provisional):

> *Humphrey, N. (2026). Pan-Cancer Hierarchical Spike-and-Slab — Replication and Structural-Prior Validation. GitHub: [URL TBD]. Zenodo: [DOI TBD].*

The replicated method:

> *Samorodnitsky, S., Hoadley, K. A., & Lock, E. F. (2022). A Hierarchical Spike-and-Slab Model for Pan-Cancer Survival Using Pan-Omic Data. BMC Bioinformatics 23(1):235.*

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License

MIT. The replication-baseline scripts in `scripts/replication/` are patched verbatim from the upstream repository. Patches to upstream files are documented inline in source. See `LICENSE`.